LaTeX 101: Level Up Your Lab Reports

Your mission: build a lab portfolio that formats itself, and keeps growing all semester

By Preethi Siva Kumar and Dr. Kartik Bulusu
MAE 1001 - Fall 2026

Why LaTeX? Housekeeper vs. Architect

Word = Housekeeper

- You format every heading, figure and number by hand
- Move one picture and the whole page shifts
- Figure numbers break when things move

LaTeX = Architect

- You write the plan; the computer builds the layout
- Figures, tables and equations number themselves
- Contents and references update on every recompile



Today's one rule: you write the content, and LaTeX handles the looks. Never fix fonts or margins by hand.

Every .tex File Has Two Parts

```
\documentclass[11pt]{article}
\usepackage{graphicx}    % pictures
\usepackage{amsmath}     % equations
\title{MAE 1001 Lab Portfolio}
\author{Your Name}

\begin{document}
\maketitle
\section{Week 1: SolidWorks Part}
Your writing goes here.
\end{document}
```

1. Preamble (the blueprint)

Everything above \begin{document}. It sets the rules and loads add-ons called packages. The template already has it, so you can leave it alone.

2. Body (the content)

Everything between \begin and \end{document}. This is where you write, and it's where you'll spend all your time.

Level 0: Set Up Overleaf in 5 Minutes

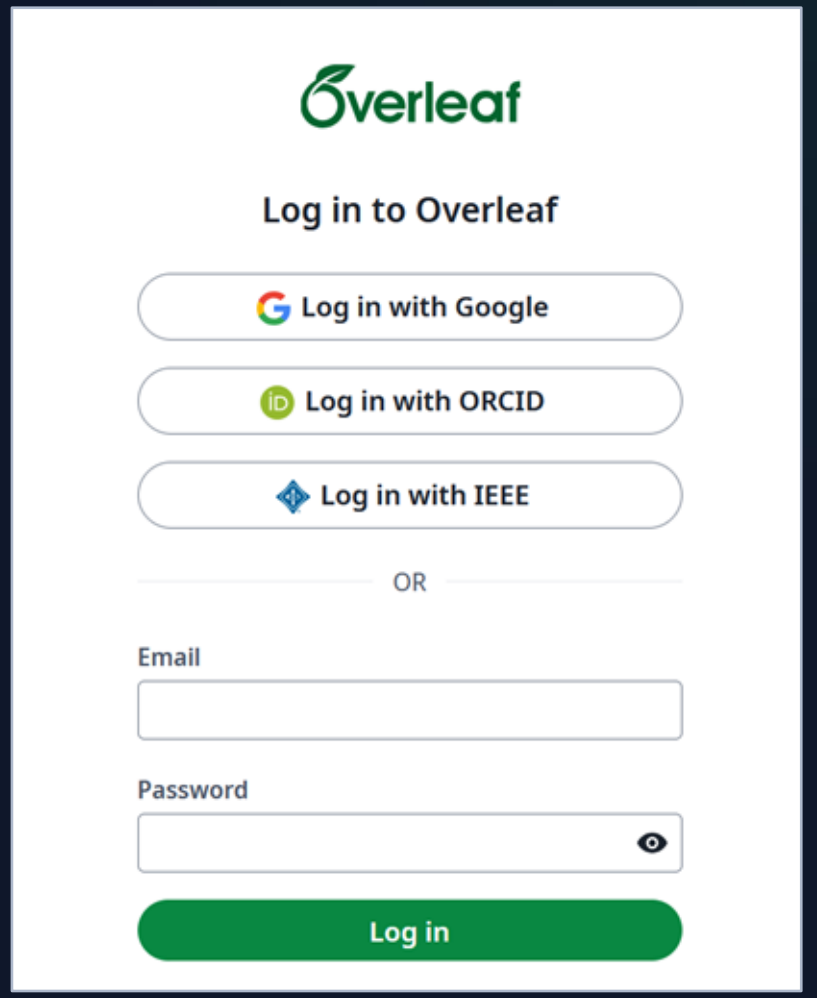
1. **Log in**
   Go to overleaf.com and sign in with your university email.

2. **Grab the template**
   Download MAE1001_LaTeX_Portfolio_Template.zip from EdStem. Don't unzip it!

3. **Upload it**
   New Project → Upload Project → pick the .zip file.

4. **Say hi**
   Open main.tex and replace the TODOs next to Name and Student ID.

5. **Hit Recompile**
   Or press Ctrl/Cmd + S. Your PDF shows up on the right. Level complete!



Your Quest Map

Every assignment is a quest. For each one, upload your pictures and write about what you did. More quests unlock as the semester goes on.

QUEST 1 · Weeks 2–3

Bias in Dice

Upload: 2 pics

Write: Guess, results table, verdict

QUEST 2 · Week 1

SolidWorks Part

Upload: 2 pics

Write: Goal, steps, reflection

QUEST 3 · Week 2

SolidWorks Assembly

Upload: 1 pic

Write: Team, mates table, challenges

QUEST 4 · Week 3

Screwdriver Notebook

Upload: 4 pics

Write: Each step, parts table, answers

QUEST 5 · Today

LaTeX vs. Word

Upload: Nothing!

Write: 3 pros, 3 cons, your pick

QUEST 6+ · LATER

Future Assignments

Copy the COPY-ME block and keep building. See slide 16.

Win condition: one PDF with every quest done, a Conclusion, auto-made Contents, and zero TODOs left.

Your Project: One File + Pictures

IN OVERLEAF'S FILE LIST

```
MAE1001-Portfolio/
├── main.tex
└── images/
    ├── w1-sketch.png
    ├── w1-part.png
    ├── w2-exploded.png
    ├── w3-dice-setup.png
    └── ...
```

main.tex = everything you write

Header, all your quests and the conclusion, top to bottom in one file.

images/ = your pictures

Upload a photo with the same name as a gray placeholder and it swaps in automatically.

TODO = your turn

Replace every TODO with your own words. Press Ctrl+F (Cmd+F) and search TODO to find them all.

Cheat Sheet: 12 Commands for 90% of Your Report

`\section{...}`

Big heading, numbered for you

`\subsection{...}`

Smaller heading

`\textbf{...}`

Bold text

`\textit{...}`

Italic text

`itemize + \item`

Bullet list

`enumerate + \item`

Numbered list

`$ ... $`

Math inside a sentence

`equation`

Numbered math on its own line

`\includegraphics`

Drop in a picture

`\caption + \label`

Name it and tag it

`\ref{...}`

Writes "Figure 3" for you

`% ...`

A note to yourself; not printed

Stuck? Keep this slide open in another tab, or ask your AI sidekick (slide 17).

Power-Up 1: Headings, Lists and Math

YOU TYPE

```
\section{SolidWorks Part}
\subsection{Objective}
I made \textbf{one-third} of a hexagon.
\subsection{Steps}
\begin{enumerate}
  \item Sketched the shape
  \item Extruded it
\end{enumerate}
Each part covers $120^\circ$.  % in a sentence
\begin{equation}              % on its own line
  \theta = \frac{360^\circ}{3} = 120^\circ
\end{equation}
```

YOU GET

2 SolidWorks Part

2.1 Objective

TODO: In 2–3 sentences, what did you make and why?

2.2 Steps

1. TODO: First step, e.g. Sketched the shape on the Front Plane
2. TODO: Second step
3. TODO: Third step

Three parts make one hexagon, so each part covers

$$\theta = \frac{360^\circ}{3} = 120^\circ \tag{2}$$

Tip: every \begin{...} needs a matching \end{...}, like opening and closing a bracket.

Power-Up 2: Adding Pictures

YOU TYPE

```
% 1. Upload your picture into images/
% 2. Point to it by name:
\begin{figure}[H]
  \centering
  \includegraphics[width=0.45\textwidth]{images/w1-sketch.png}
  \includegraphics[width=0.45\textwidth]{images/w1-part.png}
  \caption{My sketch (left) and finished part (right)}
  \label{fig:part}
\end{figure}

As shown in Figure \ref{fig:part}, ...
```

YOU GET



Figure 3: My sketch (left) and finished part (right)

Shortcut: upload a photo with the same name as a gray placeholder (like w1-part.png) and it swaps in with zero code. Two pictures in one figure sit side by side.

Power-Up 3: Tables for Your Data

YOU TYPE

```
\begin{table}[H]
  \centering
  \begin{tabular}{|c|c|c|}
    \hline
    Face & Trial 1 & Trial 2 \\
    \hline
    1    & 9       & 18      \\
    2    & 12      & 15      \\
    \hline
  \end{tabular}
  \caption{How many times each face came up}
\end{table}
```

YOU GET

Face	Trial 1	Trial 2
1	0	0
2	0	0
3	0	0
4	0	0
5	0	0
6	0	0

Table 1: How many times each face came up

Remember: & = next column, \\ = next row, \hline = a line across, and |c|c|c| = three centered columns with lines between.

Quest 1 Bias in Dice Experiment

UPLOAD

Photo of your experiment setup

`images/w3-dice-setup.png`

WRITE

- Your guess: fair or biased? Use P = 1/6
- Results table: counts per face, both trials
- Verdict: was it fair? Did Trial 2 look different?

Pro tip: Ask AI to turn your CSV into a LaTeX table. Then double-check every number!

LATEX SKILLS YOU'LL USE

`equation` `tabular` `\hline` `\ref`

FROM main.tex

```
\begin{equation}
  P = \frac{1}{6} \approx 0.167
\end{equation}

\begin{tabular}{|c|c|c|}
  \hline
  Face & Trial 1 & Trial 2 \\
  \hline
  1 & 9 & 18 \\
  \hline
\end{tabular}
```

Quest 2 SolidWorks Part

UPLOAD

Screenshot of your fully defined sketch
`images/w1-sketch.png`

Isometric view of the finished one-third hexagon
`images/w1-part.png`

WRITE

- Objective: what you made and why it's one-third of a hexagon
- Your steps as a numbered list
- Reflection: what was hard, and what would you change?

Pro tip: Crop screenshots to just the part. Lots of empty gray space makes it look tiny.

LATEX SKILLS YOU'LL USE

`\section` `enumerate` `equation` `\ref`

FROM main.tex

```
\subsection{Steps}
\begin{enumerate}
  \item Sketched the shape
  \item Extruded it 2 mm
\end{enumerate}
\begin{equation}
\theta=\frac{360^\circ}{3}=120^\circ
\end{equation}
As shown in Figure \ref{fig:part}, ...
```

Quest 3 SolidWorks Assembly

UPLOAD

A view of your team's assembled hexagon

`images/w2-exploded.png`

WRITE

- Team roles: who designed which part
- Table of every mate (type and what it connects)
- Challenges, and how you used Pack and Go

Pro tip: Fill in the mates table with the assembly open, not from memory.

LATEX SKILLS YOU'LL USE

`itemize` `\textbf` `tabular`

FROM main.tex

```
\begin{itemize}
  \item \textbf{Alex} designed part A
\end{itemize}

\begin{tabular}{|c|c|c|}
  \hline
  Mate & Type & What it connects \\
  \hline
  1 & Coincident & Edge A to B \\
  \hline
\end{tabular}
```

Quest 4 Screwdriver Lab Notebook

UPLOAD

One photo at every dissection step (4 in total)

`images/w3-step1.png … w3-step4.png`

WRITE

- Each step as a numbered list: what came off and which tool you used
- Parts table: name, material, job
- Answers to all the lab notebook questions

Pro tip: Shoot every step on the same background so your photos line up nicely.

LATEX SKILLS YOU'LL USE

`enumerate` `4 pictures` `tabular`

FROM main.tex

```
\begin{enumerate}
  \item Pulled off the battery
  \item Unscrewed the shaft
\end{enumerate}

% 4 pictures at 0.45 width
% = 2 per row, automatically
\includegraphics[width=0.45\textwidth]
  {images/w3-step1.png}
```

Quest 5 LaTeX vs. Word

UPLOAD

Nothing to upload. This one is all you!

WRITE

- At least 3 pros and 3 cons of LaTeX compared to Word
- Use what you noticed today, not a list from the internet
- One sentence: which will you use for your next lab report, and why?

Pro tip: Think about equations, where pictures landed, and how long things took you.

LATEX SKILLS YOU'LL USE

`tabular` `p{7cm}` `\hline`

FROM main.tex

```
% p{7cm} = 7 cm column, text wraps
\begin{tabular}{|p{7cm}|p{7cm}|}
  \hline
  \textbf{Pros of LaTeX} &
  \textbf{Cons of LaTeX} \\
  \hline
  Equations look great & TODO \\
  \hline
\end{tabular}
```

Quest 6+: Add Any Future Assignment in 3 Steps

1 Copy

Near the bottom of main.tex, find the COPY-ME block. Select it and copy it.

```
% \section{Week X: ...}
```

2 Paste

Paste it just above the LaTeX vs. Word section. Keep it selected and press this to remove the % signs:

```
Ctrl + /
(Mac: Cmd + /)
```

3 Rename

Change the title and the picture name, then Recompile. It joins the Contents automatically!

```
\section{Arduino Lab}
```

Picture names: start with the week so nothing clashes, like w4-circuit.png. Upload it to images/, then swap wX-photo.png in your new block for that name.

Your AI Sidekick: Gemini, ChatGPT or Claude

AI is great at LaTeX grunt work. Try these prompts (you supply the content; AI helps with the code):

Fix an error

"Overleaf says 'Missing $ inserted' on line 14. Here's my code: [paste]. Explain the mistake and show the fixed line."

Turn data into a table

"Turn this CSV into a simple LaTeX table with \hline lines and a caption: [paste]"

DO

- Paste one section at a time
- Recompile after every change
- Check every number and equation yourself

Write an equation

"Write this as a numbered LaTeX equation: kinetic energy equals one half I omega squared."

Polish your writing

"Suggest clearer wording for this paragraph. Keep my ideas and don't add new facts: [paste]"

DON'T

- Let AI write your results or reflections
- Trust packages or citations it invents
- Use AI according to the course policy

Free Loot: Header, Contents and Figure List

ALREADY IN main.tex

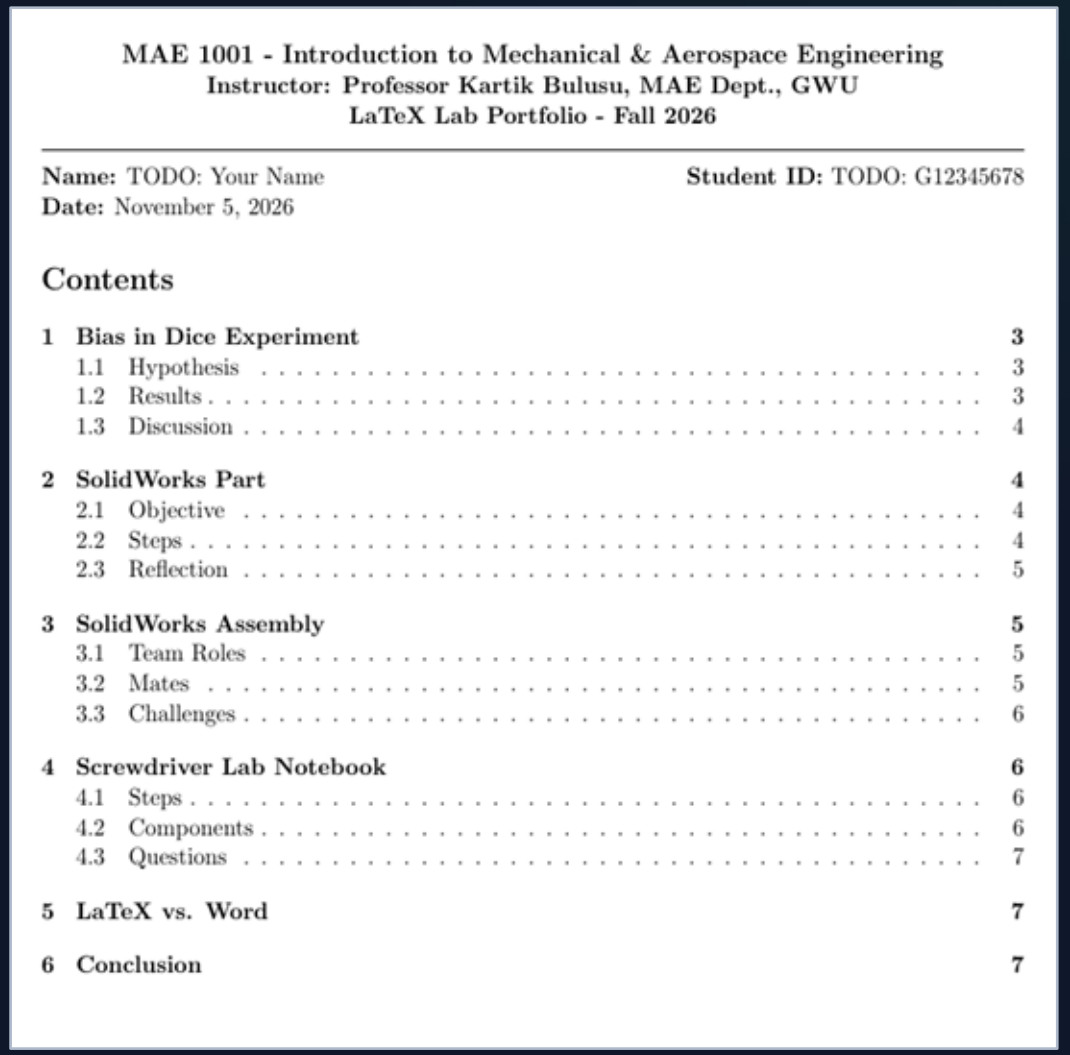
```
\textbf{Name:} Alex Smith \hfill
\textbf{Student ID:} G12345678

\tableofcontents   % from your \section{}s
\listoffigures     % from your \caption{}s
```

See ?? instead of a number? Recompile again.

Put \label after \caption so the number is right.

Write Figure \ref{...} and never type numbers by hand.



MAE 1001 - Introduction to Mechanical & Aerospace Engineering
Instructor: Professor Kartik Bulusu, MAE Dept., GWU
LaTeX Lab Portfolio - Fall 2026

Name: TODO: Your Name **Student ID:** TODO: G12345678
Date: November 5, 2026

Contents

1 Bias in Dice Experiment 3
1.1 Hypothesis . . . 3
1.2 Results . . . 3
1.3 Discussion . . . 4

2 SolidWorks Part 4
2.1 Objective . . . 4
2.2 Steps . . . 4
2.3 Reflection . . . 5

3 SolidWorks Assembly 5
3.1 Team Roles . . . 5
3.2 Mates . . . 5
3.3 Challenges . . . 6

4 Screwdriver Lab Notebook 6
4.1 Steps . . . 6
4.2 Components . . . 6
4.3 Questions . . . 7

5 LaTeX vs. Word 7

6 Conclusion 7

Bug Squashing: 5 Errors You'll Probably See

Error message	What it means	Fix
`Missing $ inserted`	You used _ or ^ outside math	Wrap it in $...$, e.g. K_{rot}
`File `w1-part.png' not found`	Name or folder doesn't match	Check spelling and capitals; is it in images/?
`Undefined control sequence`	Typo in a command	Check the line number in the log
`Figure ?? in the PDF`	Label is missing or not updated yet	\label after \caption, then recompile
`% or & disappears or errors`	These are special characters	Type \% \& \# _ instead



Errors are normal, so don't panic! Click the red number next to Recompile and fix the FIRST error only. Most of the others disappear with it.

Word vs. LaTeX: How Does Your Table Compare?

	Microsoft Word	LaTeX (Overleaf)
Pros	+ Easy to start: what you see is what you get + Great for short memos, letters and forms + Track changes and comments everyone knows	+ Beautiful equations with a few keystrokes + Numbering, contents and references are automatic + Same clean layout every time, even at 50 pages + Free and online, with live team editing in Overleaf
Cons	– Equations are slow and fiddly – Numbers and references break as things move – Pictures jump around in long reports	– Commands to learn at first – Error messages can look scary – Must recompile to see changes – Overkill for a quick one-page flyer



Verdict: use Word for quick one-pagers, and LaTeX for long reports with math, figures and references, like your portfolio.

Save Point: Download Your PDF

1. **Recompile one last time**

   No red errors? No TODOs left? Good to go.

2. **Click Download PDF**

   It's the download arrow at the top of the PDF preview. Hover over it to see "Download PDF".

3. **Find main.pdf**

   It saves to your Downloads folder as main.pdf.

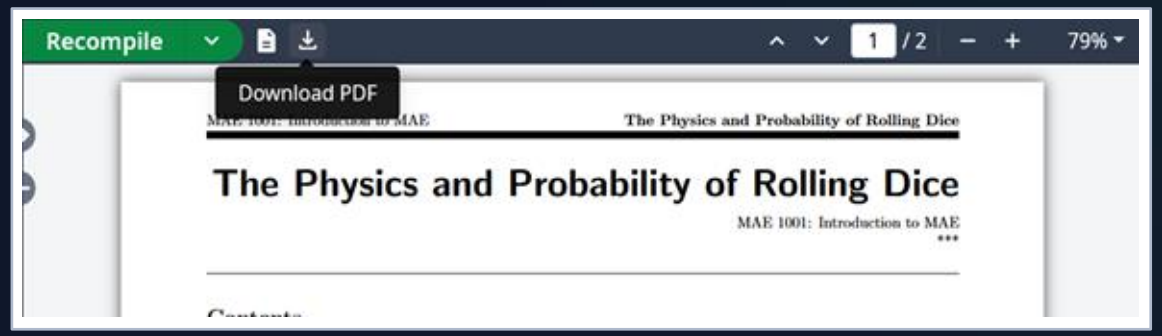
4. **Rename it**

   Add your name, e.g. Smith_Alex_LaTeX_Portfolio.pdf

5. **Open it, then submit**

   Flip through: pictures, tables and Contents all there? Upload it to the assignment.

WHERE TO CLICK



Always open before you submit. A gray REPLACE ME picture or a stray TODO is easy to spot in the PDF and easy to miss in the code.

Final Boss: Checklist and XP

BEFORE YOU SUBMIT

- Name and student ID are in the header
- All 5 quests plus the Conclusion are done
- Every gray REPLACE ME picture is now yours
- Zero TODOs left (search with Ctrl+F)
- Every figure and table is cited with \ref
- It compiles with no red errors
- You downloaded the PDF (icon next to Recompile)

Criterion	What we look for	XP
Structure	Sections, subsections, Contents, lists	20
Pictures	All required, captioned, labeled and cited	25
Tables	Dice, mates, parts and pros/cons tables	15
Equations	Hexagon and fair-die equations, inline math	15
Writing	Clear, your own words, every TODO answered	20
Clean build	No errors; PDF submitted	5
Total		100